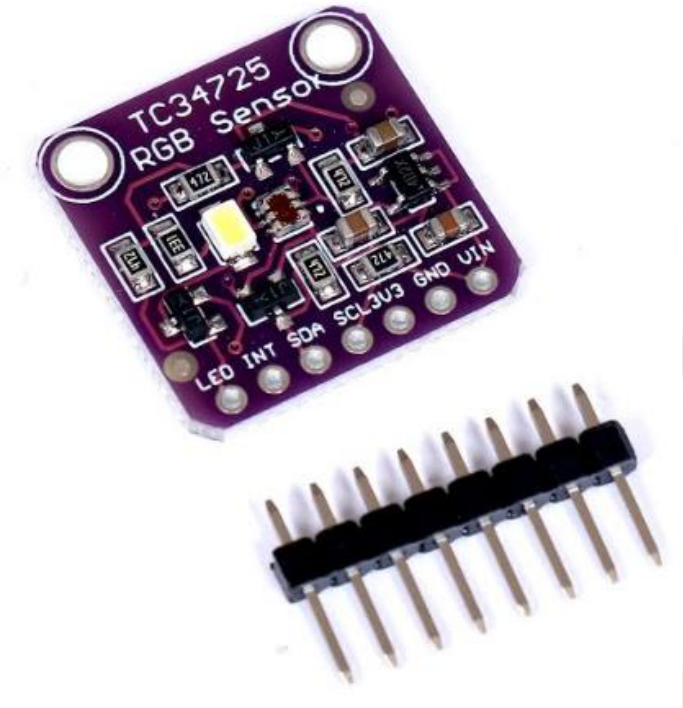
CJMCU-34725 TCS34725 Color Sensor RGB Module



The CJMCU-34725 TCS34725 Color Sensor RGB Module provides digital return values for red, green, blue (RGB) and bright light sensing digital Mingguang induction return value. Color-sensitive photodiode integrated cut and localized infrared shading filter, to minimize the incident infrared spectrum components.

So that color management more accurate. It has high sensitivity, wide dynamic range, and IR shading filters. The TCS3472 ideal for color-sensitive component solutions under varying light conditions and through attenuating materials.

An IR blocking filter, integrated on-chip and localized to the color sensing photodiodes, minimize the IR spectral component of the incoming light and allows color measurements to be made accurately. The filter means you'll get much truer color than most sensors since humans don't see IR. The sensor also has an incredible 3,800,000:1 dynamic range with adjustable integration time and gain so it is suited for use behind darkened glass.

High sensitivity, wide dynamic range and an infrared light blocking filter makes TCS3472 become ideal lighting conditions change and through color sensitive element attenuating materials under conditions of a solution.

FEATURES:

- The TCS3472 device provides digital return values for red, green, blue (RGB) and bright light sensing
- Color-sensitive photodiode integrated cut and localized infrared shading filter, to minimize the incident infrared spectrum components, so that color management more accurate.
- The high sensitivity.
- Wide dynamic range.
- IR shading filters make the TCS3472 ideal for color-sensitive component solutions under varying light conditions and through attenuating materials.

SPECIFICATIONS:

- Model: TCS34725
- Operating Voltage (VDC): 3.3
- Supply Voltage(VDC): 3.3 to 5
- PCB Thickness(mm): 1.7
- Dimension(LxWxH): 20.5x20.5x3
- Weight(gm): 1 (approx)

FUNCTIONAL DESCRIPTION:

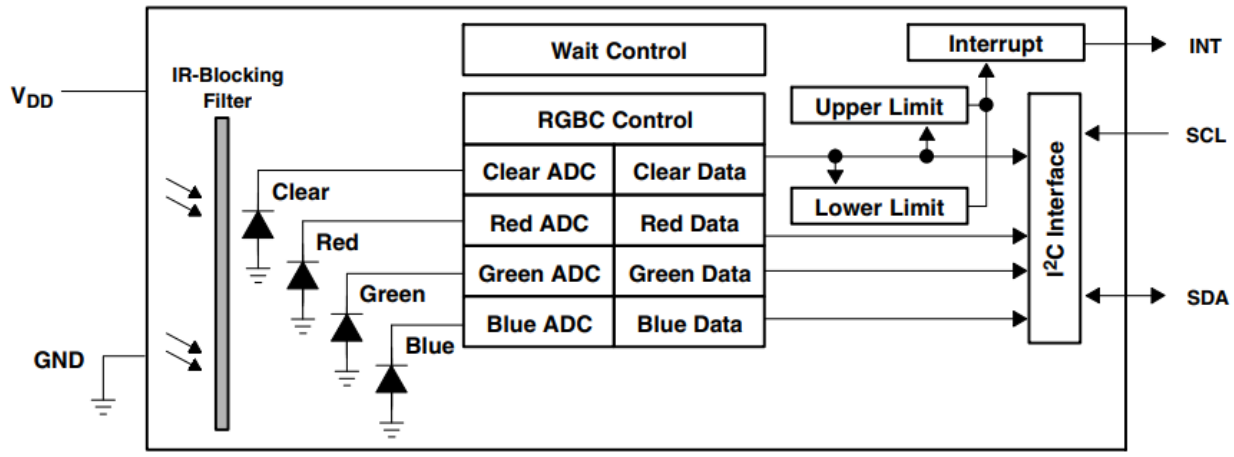


Figure1. Block diagram of TCS3472 light-to-digital converter

- The TCS3472 light-to-digital converter contains a 3 × 4 photodiode array, four analog-to-digital converters (ADC) that integrate the photodiode current, data registers, a state

machine, and an I2C interface. The 3 × 4 photodiode array is composed of red-filtered, green-filtered, blue-filtered, and clear (unfiltered) photodiodes. In addition, the photodiodes are coated with an IR-blocking filter.

- The four integrating ADCs simultaneously convert the amplified photodiode currents to a 16-bit digital value. Upon completion of a conversion cycle, the results are transferred to the data registers, which are double-buffered to ensure the integrity of the data. All of the internal timing, as well as the low-power wait state, is controlled by the state machine. Communication of the TCS3472 data is accomplished over a fast, up to 400 kHz, two-wire I2C serial bus. The industry standard I2C bus facilitates easy, direct connection to microcontrollers and embedded processors.
- In addition to the I2C bus, the TCS3472 provides a separate interrupt signal output. When interrupts are enabled, and user-defined thresholds are exceeded, the active-low interrupt is asserted and remains asserted until it is cleared by the controller.
- This interrupt feature simplifies and improves the efficiency of the system software by eliminating the need to poll the TCS3472. The user can define the upper and lower interrupt thresholds and apply an interrupt persistence filter. The interrupt persistence filter allows the user to define the number of consecutive out-of-threshold events necessary before generating an interrupt. The interrupt output is open-drain, so it can be wire-ORed with other devices.

PRINCIPLE OF OPERATION:

- **System States:**

An internal state machine provides system control of the RGBC and power management features of the device. At power up, an internal power-on-reset initializes the device and puts it in a low-power Sleep state. When a start condition is detected on the I2C bus, the device transitions to the Idle state where it checks the Enable Register (0x00) PON bit. If PON is disabled, the device will return to the Sleep state to save power. Otherwise, the device will remain in the Idle state until the RGBC function is enabled (AEN). Once enabled, the device will execute the Wait and RGBC states in sequence as indicated in Figure 2. Upon completion and return to Idle, the device will automatically begin a new Wait-RGBC cycle as long as PON and AEN remain enabled.

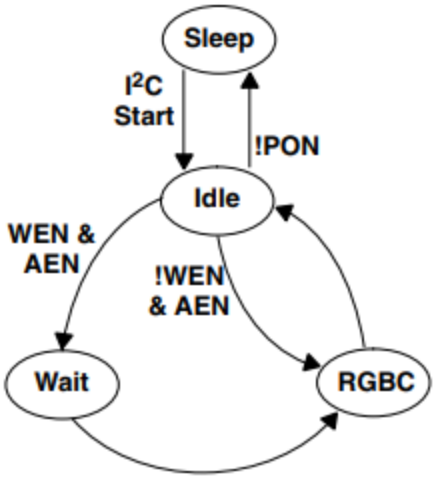


Figure 2. Simplified State Diagram

- **RGBC Operation:**

The RGBC engine contains RGBC gain control (AGAIN) and four integrating analog-to-digital converters (ADC) for the RGBC photodiodes. The RGBC integration time (ATIME) impacts both the resolution and the sensitivity of the RGBC reading. Integration of all four channels occurs simultaneously and upon completion of the conversion cycle, the results are transferred to the color data registers. This data is also referred to as channel count. The transfers are double-buffered to ensure that invalid data is not read during the transfer. After the transfer, the device automatically moves to the next state in accordance with the configured state machine.

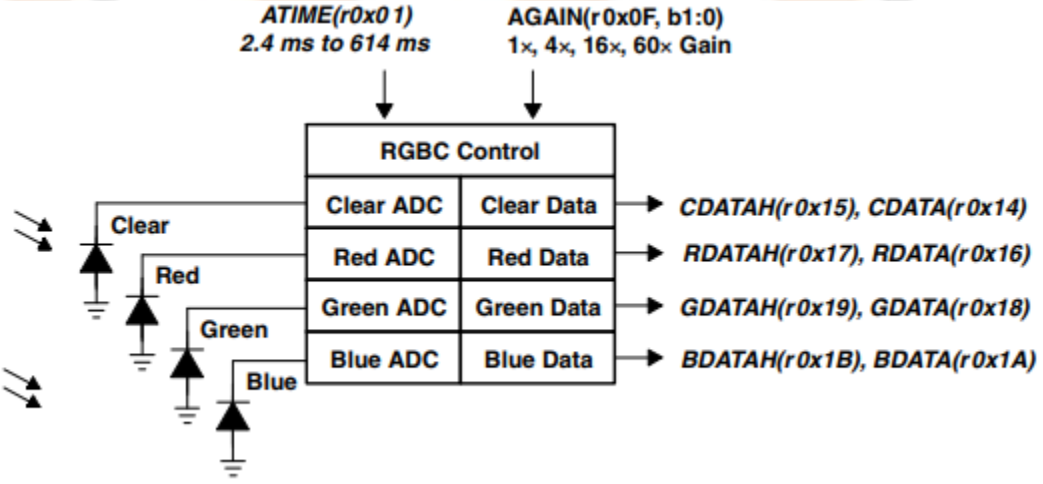


Figure 3. RGBC Operation

- **Interrupts:**

The interrupt feature simplifies and improves system efficiency by eliminating the need to poll the sensor for light intensity values outside of a user-defined range. While the interrupt function is always enabled and its status is available in the status register (0x13), the output of the interrupt state can be enabled using the RGBC interrupt enable (AIEN) field in the enable register (0x00). Two 16-bit interrupt threshold registers allow the user to set limits below and above a desired light level. An interrupt can be generated when the Clear data (CDATA) is less than the Clear interrupt low threshold (AILTx) or is greater than the Clear interrupt high threshold (AIHTx). It is important to note that the thresholds are evaluated in sequence, first the low threshold, then the high threshold. As a result, if the low threshold is set above the high threshold, the high threshold is ignored and only the low threshold is evaluated. To further control when an interrupt occurs, the device provides a persistence filter. The persistence filter allows the user to specify the number of consecutive out-of-range Clear occurrences before an interrupt is generated. The persistence filter register (0x0C) allows the user to set the Clear persistence filter (APERS) value. See the persistence filter register for details on the persistence filter value. Once the persistence filter generates an interrupt, it will continue until a special function interrupt clear command is received.

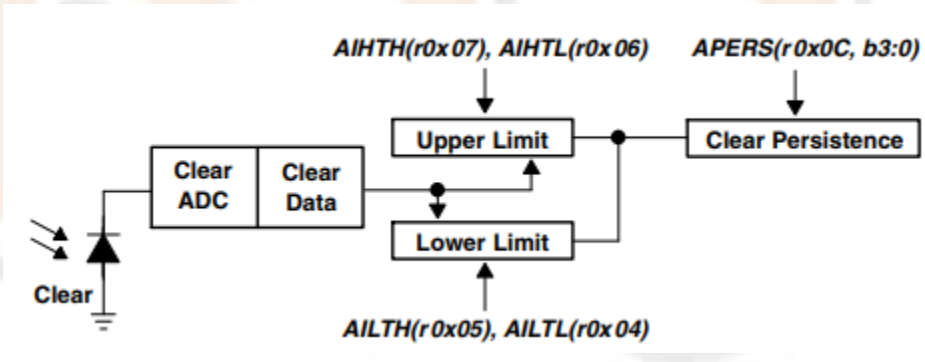


Figure 4. Programmable Interrupt

- **System Timing:**

The system state machine shown in Figure 2 provides an overview of the states and state transitions that provide system control of the device. This section highlights the programmable features, which affect the state machine cycle time, and provides details to determine system level timing. When the power management feature is enabled (WEN), the state machine will

transition to the Wait state. The wait time is determined by WLONG, which extends normal operation by 12× when asserted, and WTIME. The formula to determine the wait time is given in the box associated with the Wait state in Figure 5. When the RGBC feature is enabled (AEN), the state machine will transition through the RGBC Init and RGBC ADC states. The RGBC Init state takes 2.4 ms, while the RGBC ADC time is dependent on the integration time (ATIME). The formula to determine RGBC ADC time is given in the associated box in Figure 5. If an interrupt is generated as a result of the RGBC cycle, it will be asserted at the end of the RGBC ADC.

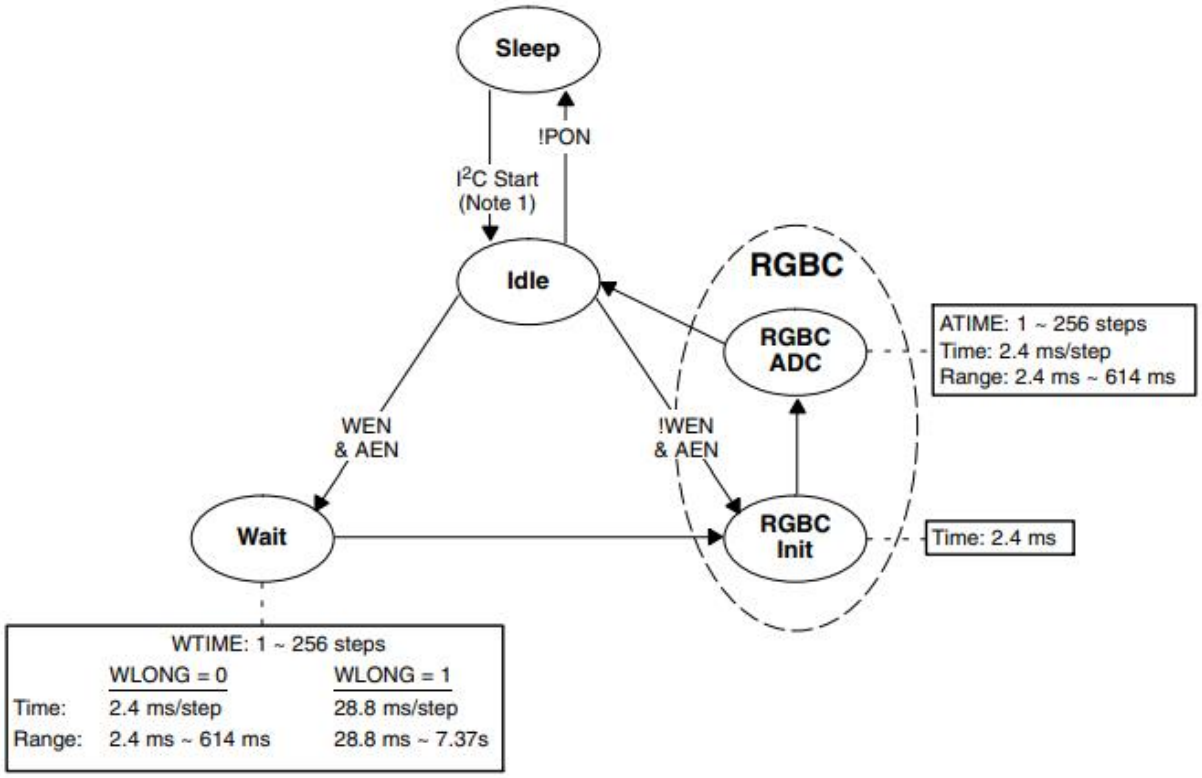


Figure 5. Detailed State Diagram

- **Power Management:**

Power consumption can be managed with the Wait state, because the Wait state typically consumes only 65 µA of IDD current.

- **I2C Protocol:**

Interface and control are accomplished through an I2C serial compatible interface (standard or fast mode) to a set of registers that provide access to device control functions and output data. The devices support the 7-bit I 2C addressing protocol. The I2C standard provides for three types of bus transaction: read, write, and a combined protocol (Figure 6). During a write operation, the first byte written is a command byte followed by data. In a combined protocol, the first byte written is the command byte followed by reading a series of bytes. If a read command is issued, the register address from the previous command will be used for data access. Likewise, if the MSB of the command is not set, the device will write a series of bytes at the address stored in the last valid command with a register address. The command byte contains either control information or a 5-bit register address. The control commands can also be used to clear interrupts.

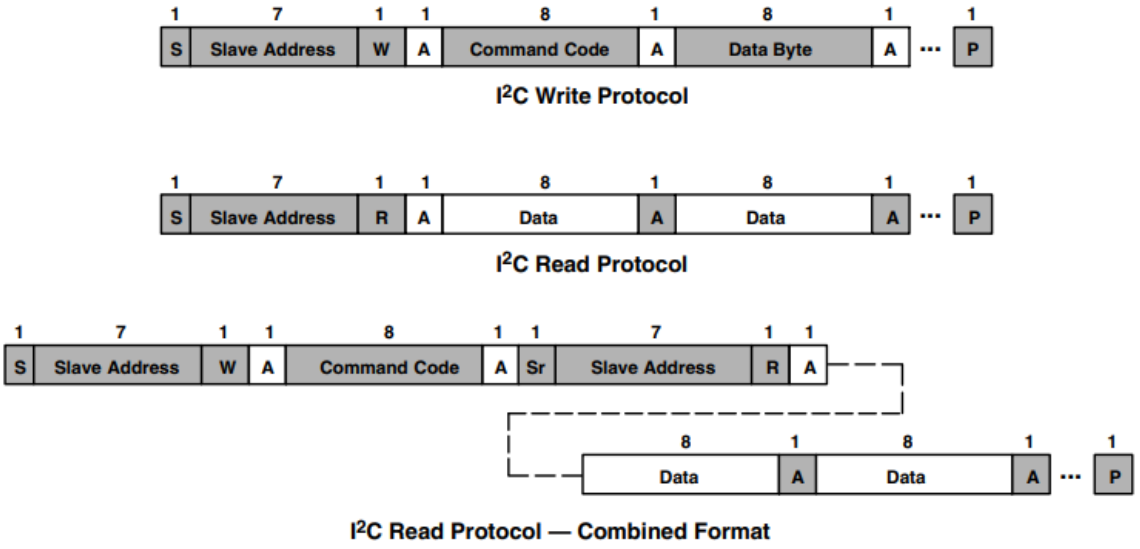


Figure 6. I2C Protocols

APPLICATIONS:

- RGB LED Backlight Control
- Light Color Temperature Measurement
- Ambient Light Sensing for Display Backlight Control

- Fluid and Gas Analysis
- Product Color Verification and Sorting

PACKAGE INCLUDES:

- 1 x CJMCU-34725 TCS34725 Color Sensor RGB Development Board Module.
- 1 x 7*1 Burg connector Pins(Straight) (Unsoldered).